

REPTILE BIOMIMICRY

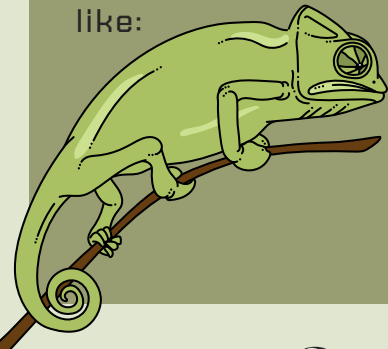
SNIGDHA GURRAM
SEASIDE SUSTAINABILITY

Chameleons

Color Changing

- Choleric Liquid Crystals (CLCs)
- Chromatophores is what lets chameleons change colors but this is different from technology made color changing

Inspiration for color changing products like:



- thermal color changing
- inspired making clothes, bags, and shoes

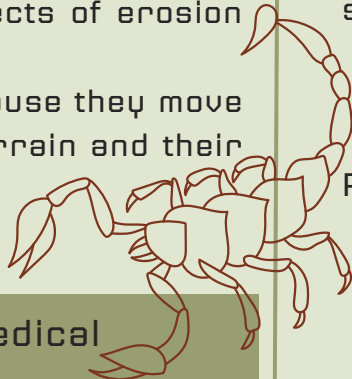
Scorpion

Behavioral

- central pattern generators

Inspiration for the robot scorpion which is being developed by the Defence Advanced Research Projects Agency (DARPA) in the United States.

- scientists looked at the exoskeleton of a scorpion to reduce the effects of erosion on steel.
- they choose scorpions because they move and survive in tough hot terrain and their reflexes are simple



Otto Schmitt started biomedical engineering with the term “biomimetics” when he got inspired by a squids’ nervous system and invented the Schmitt trigger

Biomimicry is when people learn from and mimic the strategies living organisms used to replicate during human development of projects.

Gecko

Adhesion

- dry adhesives

Inspiration for wall climbing products like:

- Sticky-bot: robot with four legs with the ability to climb smooth surfaces.
- Geckskin was invented in 2012 by Al Crosby and Duncan J. Irschick which has the ability to stick to many different types of surfaces.

Snakes

Sensing system, and balancing system

- shape memory alloys (SMAs), Piezoelectric materials, and Electroactive polymers

Pit organs of the snake inspired

- mini-viper model: designed to protect soldiers from explosives and ambush.

Sources:

- [Reptile Biomimicry](#)
- [History of Biomimicry and Geckskin](#)
- [What is Biomimicry](#)

